

GEA1000 - Tutorial 1

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Tutorial Information

- ① James (jamestwj@nus.edu.sg)
- ② 4 tutorial sessions + 1 final project presentation.
- ③ Assessment:
 - ① **Weekly Quizzes.**
 - ② **Class Part + Attendance + In-Class Quiz (ICQ)**
 - ③ **Group Project + Presentation.** (Will be discussed later on.)
 - ④ **Final Exam.**

For exact weightage of each component, do check the module canvas page.

I welcome students to drop me an email to ask questions at anytime!

What to prepare before each class

- ① Lessons will start around 5 minutes late to allow students to come in.
- ② Students are not expected to have attempted the tutorial prior coming to class. Time will be given to read and discuss.
- ③ Review the lecture videos before class.
 - ① I will review important concepts covered in the lecture videos (but not all).
- ④ Your own device, with Internet access and Excel installed.
 - ① To make things easier, I will cover mostly the usage of Excel in our classes, unless in some cases where Radiant is much preferred.

Some FAQs

Q: What if I am unable to attend lesson?

A: Drop me an email to arrange for another makeup class. For EW-classes, do try to inform me ASAP as classes for the specific tutorial end on even weeks.

Q: Will we be tested on software usage? E.g. Excel and/or Radiant.

A: Yes, basic usage on Excel will be tested in your quizzes, project and final exam.

Q: I think I require some special arrangements/ assistance for this module.

A: Up to your comfort level, kindly inform the OSA and me if you have any concerns so that I can take note for you during class. :)

Q: Will the workload for this module be heavy?

A: The workload is tedious but not heavy (imo). Else, just SU it. :)

Tutorial 1

Preparation

In this tutorial, we will be exploring a dataset containing box office records and ratings for movies. First, access the files **movies.xlsx** and tutorial 1.

The dataset contains 6000+ different data points. Take about 5 minutes to browse the dataset and Tutorial 1.

Explanation of Terminology

- ① **Title, Duration, MPAA Rating** and **Genre** are self-explanatory.
- ② **ProductionBudget**: Cost of making the movie while **WorldwideGross**: Revenue generated by the public screenings of the movie.
- ③ **DomesticGross / DomesticGrossOpening**: Former refers to revenue generated from screening in the US. Latter refers to the revenue generated from screening in the US during the opening weekend.
- ④ **IMDbRating**: A rating computed and given by various voters to the movie. It has a scale of 1 to 10, which the higher the rating, the better the movie is perceived.
- ⑤ **VoterNumbers** is the number of people voted on the IMDb rating of the movie.

Q1a)

- ① **Numerical Variable:** Can be **measured/ quantified** with a numerical value.
- ② **Categorical Variable:** Can be divided into subgroups/ categories, **typically sharing the same characteristics.**

Numerical	Categorical
ProductionBudget	Genre
WorldwideGross	DateOfRelease
DomesticGross	MPAARating
Duration	
VoterNumbers	
DomesticGrossOpening	

Title belongs to neither because there can only be one movie belonging to each title. It is a **Identifier Variable**.

PART I: IMDb - Categorical or Numerical?

Question: Is IMDb considered Categorical or Numerical?

A few fair points can be made:

- 1 The IMDb is firstly, a **number** and secondly, derived from performing **mathematical algorithms**. Hence, it should be a **numerical variable**.
- 2 However, the IMDb is based on **opinions**, which is not precisely 'measurable'. Consider the following:
 - 1 A movie-goer rates a movie 4/10 and another 8/10. Do we say that the 8/10 movie is 'twice' as good as the 4/10 movie?
 - 2 Two movie-goers rated the same movie 3/10 and 8/10 respectively. Who's opinion should we believe?

IMDb - Categorical or Numerical?

The long answer is that the IMDb-rating is technically not a numerical variable but is **treated like one**. In fact, it is considered an **ordinal variable**.

Definition

Ordinal Variables: Have a clear **hierarchy/ranking**, but the 'distance' between each ranks isn't necessarily fully quantifiable/measurable. It is considered a type of categorical variable.

Some examples of ordinal variables

- 1 Likert Scale (SD/D/N/A/SA, How happy are you on a scale from 1 to 10, etc...)
- 2 Education Level (Primary, Secondary, Pre Tertiary, etc...)

Q1b)

To count the number of blank/Unknown cells in the variable, say 'Title', we use the following commands:

Command	Purpose
=COUNTBLANK(A2: A6339)	Counts # of blanks in the specified region.
=COUNTIF(A2:A6339, "Unknown")	Count # of cells with "Unknown" in the specified region.

SUM		:	✖	✓	<i>f_x</i>	=COUNTBLANK(A2:A6339)+COUNTIF(A2:A6339, "Unknown")	
	A	L	M	N	O		
1	Title		Title				
2	Star Wars Ep. VII: The Force Awakens		"Unknown")				
3	Spider-Man: No Way Home						
4	The Avengers						
5	Top Gun: Maverick						
6	Star Wars Ep. VIII: The Last Jedi						
7	Jurassic World: Fallen Kingdom						

Q1b)

The following table shows the final count for each respective variable.

Variable	Missing/Unknown Count
DomesticGrossOpening	1895
Duration	540
MPAA Rating	323
IMDb Rating	315
Voter Numbers	296
Genre	122
Date of Release	112
Title	2

Remarks - Anomalous Data Points

Some comments about our output:

- 1 There are two movies labeled as 'Unknown' (Row 225 and 6089). A simple google search reveal there are indeed two movies titled 'Unknown' released in their respective years. So we do not treat these 2 entries as 'dirty' data.
- 2 There are some movies with ambiguous labelling. For e.g. "???? (CJ7)" (Row 2515). This is likely referring to the Hong Kong movie "长江七号". However, this 'dirty' data is likely due to Excel having difficulty in processing chinese characters.

Q1c)

Click into an empty cell, highlight the column (variable) to be considered. Here it is WorldWideGross (column C).

To compute, say minimum, we use the command "`=MIN()`".

ProductionBudget	WorldwideGross		
306000000	2064615817		=MIN(C:C)
200000000	1910041582		
225000000	1515100211		
170000000	1488732821		
262000000	1331635141		
170000000	1308323302		

You should get '0'.

Commands

The table below gives the rest of the required commands to compute the respective quantities:

Qty	Command	Output
Mean	=AVERAGE()	91872627.07
Standard Deviation	=STDEV.S()	181860051.7
First Quartile	=QUARTILE(, 1)	3731307.5
Third Quartile	= QUARTILE(, 3)	96057767.5
Median	=MEDIAN()	27012909.5
Minimum	= MIN() or QUARTILE(, 0)	0
Maximum	= MAX() or QUARTILE(, 4)	2899384102

Remarks on Excel

- ① 3 commonly used Excel quartile functions: QUARTILE, QUARTILE.INC and QUARTILE.EXC. We will use QUARTILE (as it is backward compatible). (QUARTILE and QUARTILE.INC gives the same output numerically.)
- ② There are 2 main SD functions:
 - ① STDEV.S is used when computing the SD of a **sample**.
 - ② STDEV.P is used when computing the SD of a **population**.

Since the data given is a sample of 6338 data points, we use STDEV.S here.

Evaluation on Summary Statistics

Observe there are several 0 values for WorldwideGross.

The definition of WorldwideGross is the revenue generated from screening in **public theatres**.

Hence it is possible revenues generated from other forms (such as streaming, limited, theatrical release, rentals, etc) are not accounted for.

Conclusion: If we are only interested in movies which had a public theatrical release, then we may safely ignore the 0 values data points.

Q1d)& e): Random Assignment

For this part, we will using a fresh dataset “movies_random_assign.xlsx”, which consists of 6322 cleaned entries, to perform **random assignment**.

Create a new column “Random Assignment”. In the first entry, we key in the command:

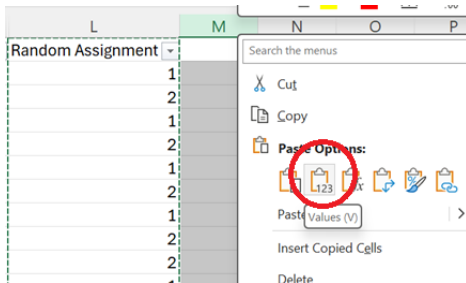
=RANDBETWEEN(1,2)

Double click on the bottom-right corner to apply to the entire column.

L2	=RANDBETWEEN(1,2)			
	A	B	C	L
1	Title	ProductionBudget	WorldwideGros	Random Assignment
2	Star Wars Ep. VII: The Force Awakens	306000000	2064615817	1
3	Spider-Man: No Way Home	200000000	1910041582	2
4	The Avengers	225000000	1515100211	1
5	Top Gun: Maverick	170000000	1488732821	2
6	Star Wars Ep. VIII: The Last Jedi	262000000	1331635141	1
7	Jurassic World: Fallen Kingdom	170000000	1308323302	2
8	Beauty and the Beast	160000000	1268697483	1
9	Aquaman	160000000	1143758700	2

Random Assignment - cont'

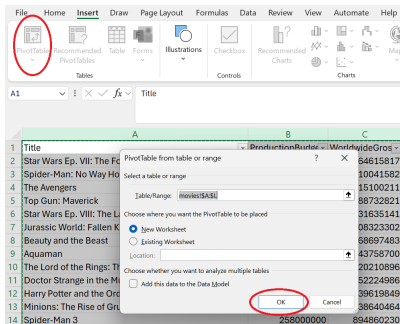
Copy the entire column, and paste it over to another column '**as values**'. (This is because Excel rand functions reruns every time you perform an action to it.)



After that, delete the old column. We will use this newly pasted column as our fixed random assigned value.

Introducing the Pivot Table

We will use the Pivot Table function in Excel to help us simplify computations. First, highlight the entire dataset. Click **Insert -> Pivot Table -> OK**. A new worksheet should pop up in your file.



Row/Column Values of Pivot Table

In your new worksheet, drag 'Random Assignment' to '**Rows**', 'Genre' to both '**Columns**' and '**Values**'. A table with rows labelled '1' and '2' and the columns labelled the various movie genres should appear (see below.)

PivotTable Fields

Choose fields to add to report:

Search

☒ Genre
☒ Random Assignment
 More tables...

Drag fields between areas below:

Drag here

Filters: Genre
 Rows: Random Assignment
 Columns: Count of Genre

You can play around with 'Column labels' to see the count of say, 'Action' movies in group 1 and 2.

Repeat the same steps but with 'Genre' and 'MPAARating' to obtain the count of various rated movies in group 1 and 2.

Random Assignment - Evaluation

We can summarize our findings and computations in the following tables:

	Action Movies	R-Rated	Count
Group 1	754	1300	3096
Group 2	820	1337	3226
Population	1574	2637	6322
	Action Movies	R-Rated	
Group 1	0.243540052	0.419897	
Group 2	0.254184749	0.414445	
Population	0.248971844	0.417115	

- 1 Observe that the proportions between group 1 and 2 are somewhat close to one another.
- 2 Moreover, both of their proportions are also close to the population proportions.

This exercise illustrates the point that allowing **randomness** in our assignment/sampling methods is likely to ensure that characteristics remain **homogeneous** in the smaller subgroups/samples.

Group Work

In your own groups, spend about 15 minutes to discuss **Section 2: Study Design**. Bear in mind the following questions:

- 1 Would you conduct an experimental or observational study? Why?
- 2 How do you obtain your sampling frame?
- 3 What are your variables? How will you measure them?
- 4 If you plan to do an experiment, will you perform any blinding? Why?
- 5 Are there any ethical considerations?

Randomized Control Trial

Our question:

"Does consuming popcorn increase the enjoyment of movies?"

- ① **Independent Variable:** Popcorn consumption ("Yes" or "No").
- ② **Dependent Variable:** Enjoyment of Movies. (Likert Scale: 1 to 5, 5 being most enjoyable)

Preliminary Idea: Have two groups of students (treatment vs control) watch the same movie, and then record their responses.

In Class Quiz (ICQ)

For the last part of the lesson, please do your ICQ:

- 1 I have emailed the class the questions and datasets for the ICQ.
- 2 Login to the module's canvas page, go to quizzes and start ICQ1.
- 3 Enter the unique ID code at the end of the quiz.
- 4 You may leave once you are done with the ICQ. :)

Please **notify me** if you have yet to receive any email from me!